GLOBAL CO_2 EMISSIONS

Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024 | Atmospheric milestone: Feb 2026

35+ Bt

annual fossil fuel & industry CO_2 today

429 ppm

atmospheric CO_2 at Mauna Loa, Feb 2026

Sources: Global Carbon Budget 2025; NASA climate vital signs; NOAA Mauna Loa

ATMOSPHERIC MILESTONE

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

NOAA Mauna Loa measurement

February 2026

~280 ppm

pre-industrial baseline

approximate level before large-scale fossil fuel use

>50%

increase above baseline

more than 1.5× pre-industrial concentration

Pre-industrial
~280 ppm

1958
Mauna Loa record begins

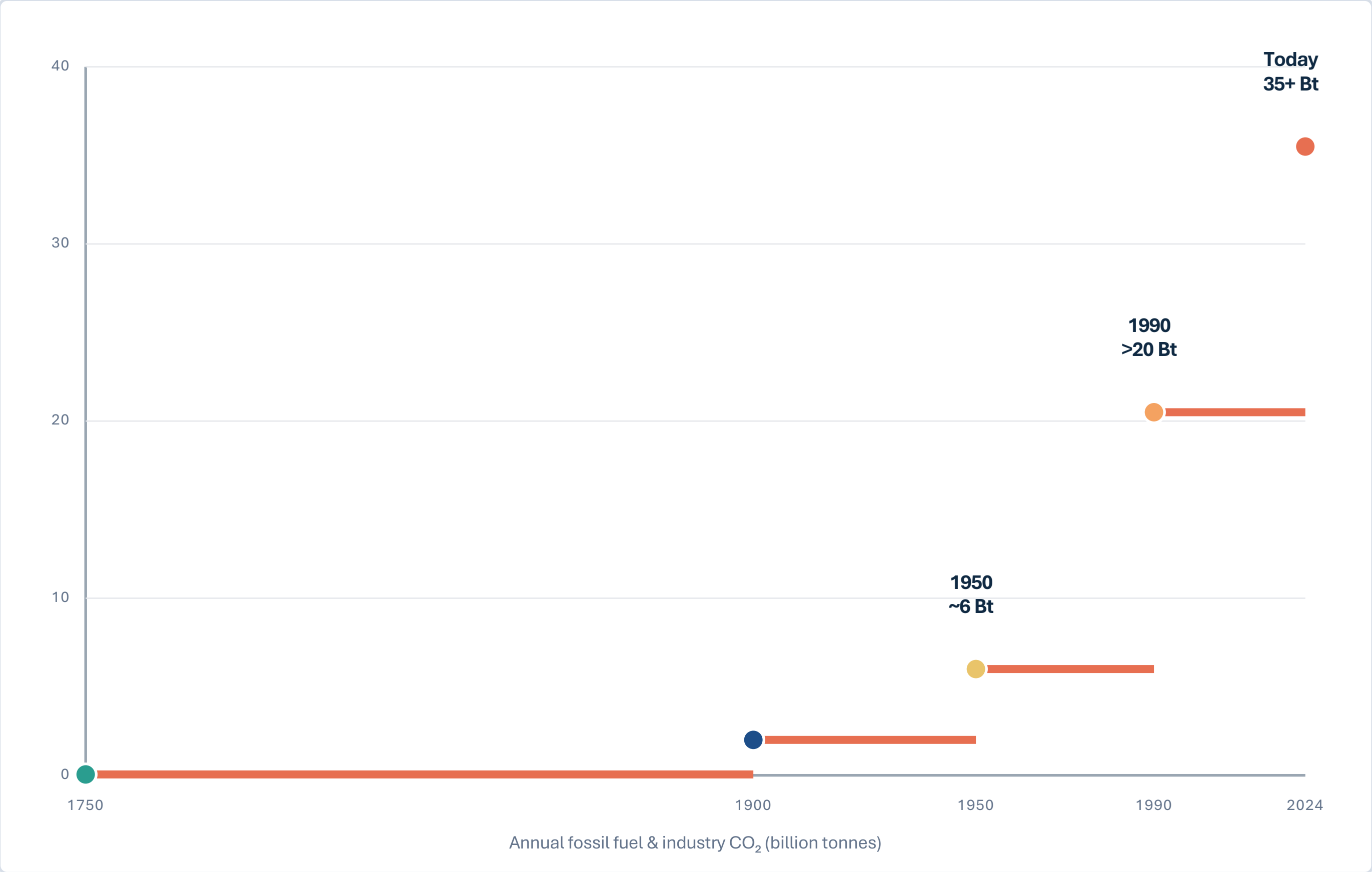
Feb 2026
429 ppm

Ground-based and satellite measurements confirm sharp acceleration, linked primarily to burning coal, oil, and natural gas.

Source: NASA climate vital signs; NOAA Mauna Loa Observatory.

GLOBAL TRAJECTORY

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



~6 Bt
world emissions in 1950

>20 Bt
world emissions by 1990

35+ Bt
today: fossil fuels and industry

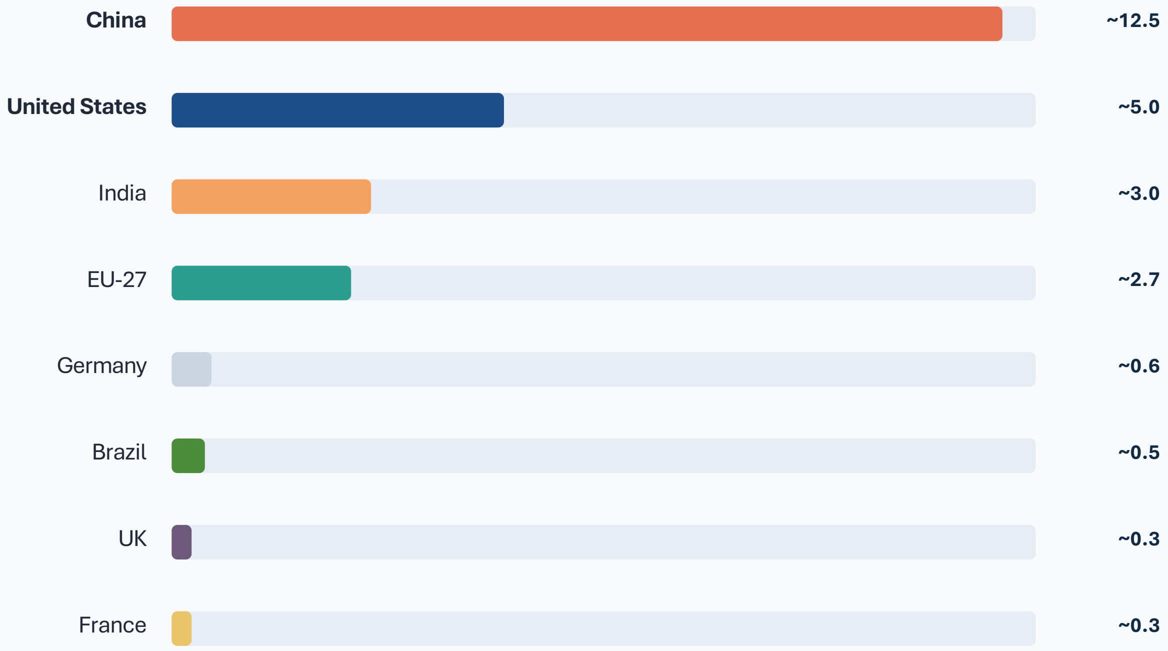
Growth remained slow until the mid-20th century. Recent growth has slowed, but global emissions have not yet peaked.

Figure reference: figure_2.jpg (trajectory)

Source: Global Carbon Budget 2025. figure_2.jpg referenced in specification; source brief provides the values shown.

COUNTRY TOTALS

China Now Emits Over 12 Billion Tonnes — More Than Double the United States



Approx. annual CO_2 emissions, billion tonnes

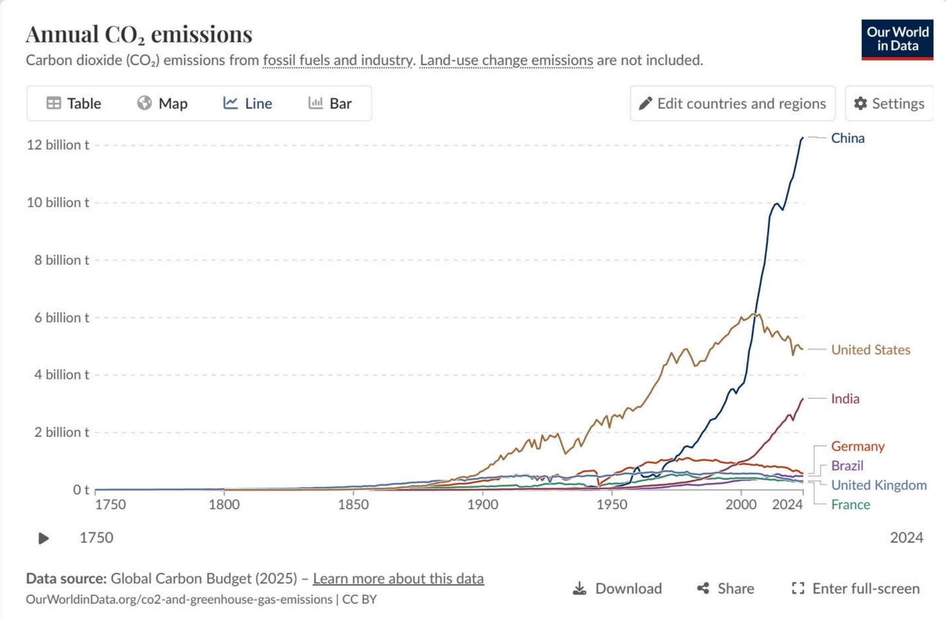


figure_3.jpg — Annual CO_2 emissions by country

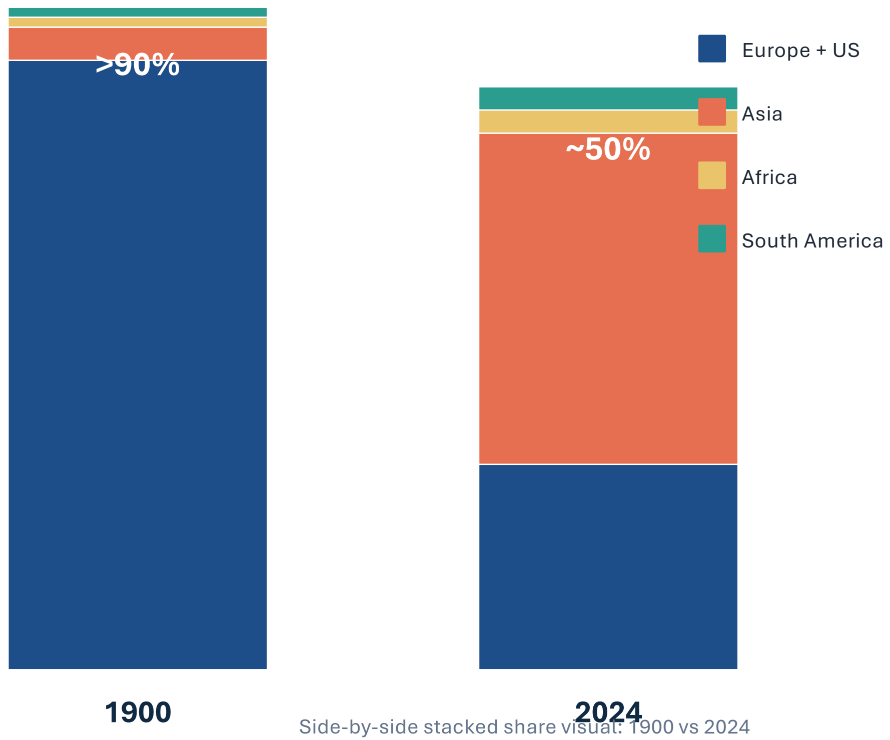
~27%
China share of global emissions

~14%
United States share

China's post-2000 rise contrasts with a US decline from its ~6 Bt peak around 2005.

Source: Global Carbon Budget 2025; Our World in Data.

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



Side-by-side stacked share visual: 1900 vs 2024

>90%
Europe + US share in 1900

>85%
Europe + US share in 1950

~50%
Asia share today

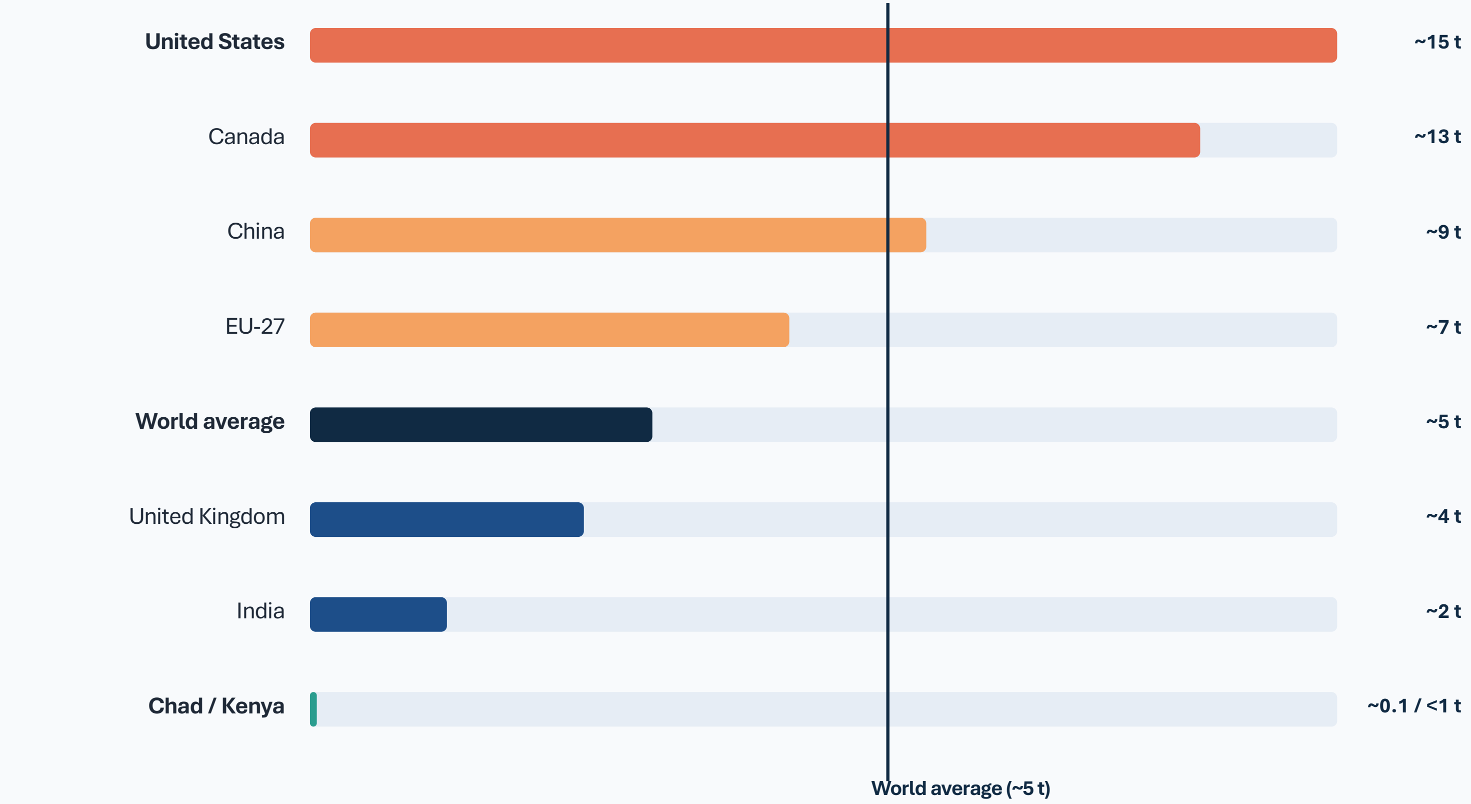
3–4%
Africa and South America each

The geographic center of emissions has shifted from the early industrial economies to Asia — home to nearly 60% of the world's population.

Source: Global Carbon Budget 2025. Shares are stated in the source brief.

PER CAPITA INEQUALITY

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



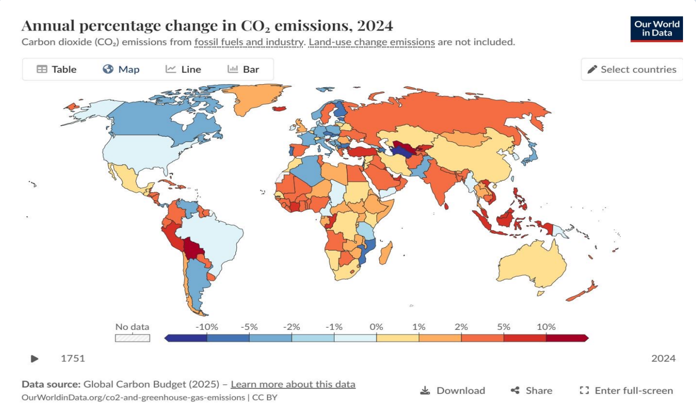


figure_1.jpg — per capita line chart

150×

gap between US / Australia and Chad / Niger

An average American or Australian emits in under two days what Chad or Niger emits in a year.

Source: Global Carbon Budget 2025; Our World in Data. Per capita values are approximate tonnes per person per year.

ANNUAL CHANGE MAP

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

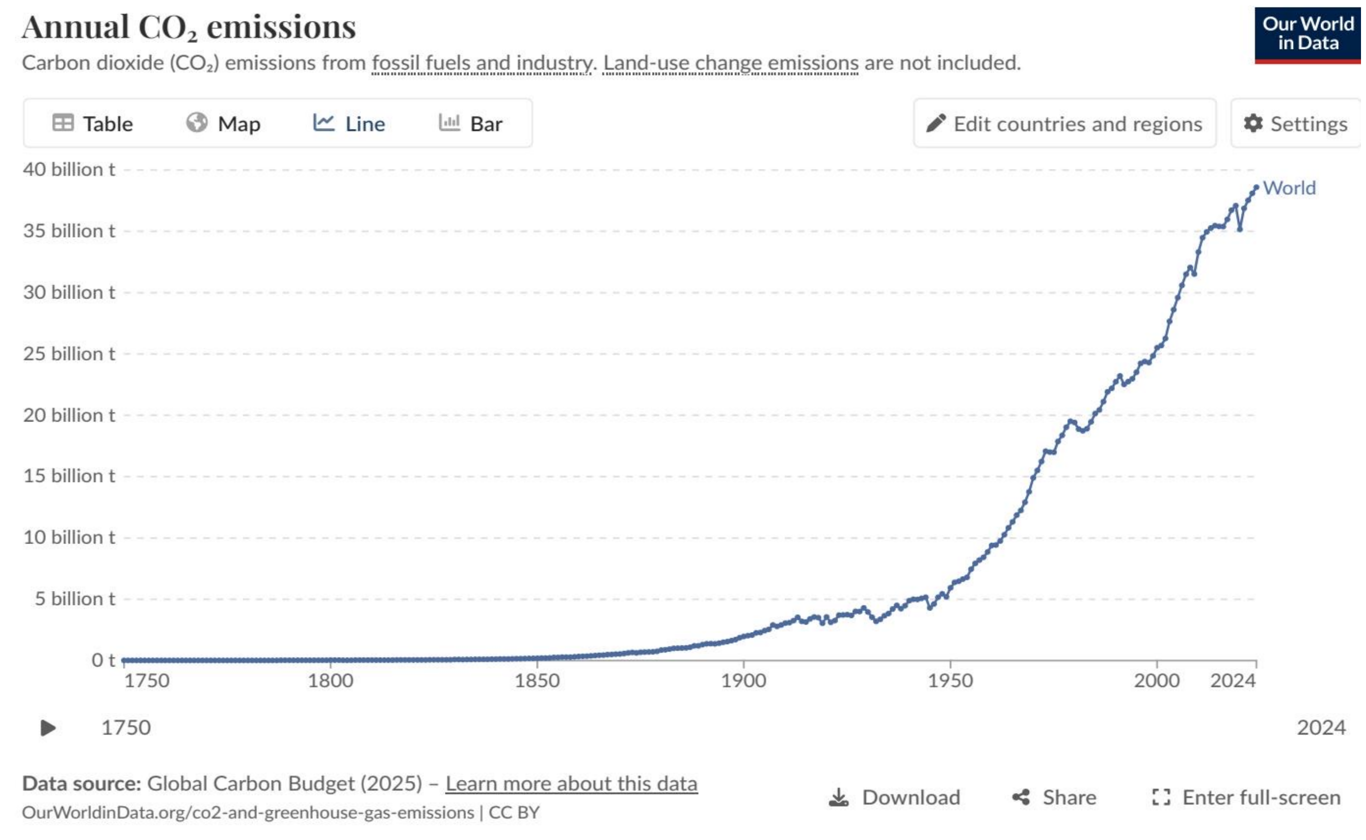


figure_4.jpg — Annual percentage change in CO_2 emissions, 2024

Reading the 2024 map

Blue — US and much of Europe: declines of roughly −1% to −5%.

Orange/red — Russia, parts of Southeast Asia, and several African nations: +2% to +10% or more.

South America — Mixed: one or two countries with very large >10% increases; others declined.

Interpretation: decarbonization progress is uneven; declining mature economies coexist with growth in parts of Eurasia, Africa, and Southeast Asia.

Source: Global Carbon Budget 2025 / Our World in Data.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have significantly different per capita emissions depending on reliance on nuclear, renewables, or fossil fuels.

Comparator	Energy-source profile stated in brief	Per capita CO_2	Policy implication
France / UK	Higher nuclear and renewable shares	**France ~4 t** **UK ~4 t**	Lower-carbon energy systems can reduce per capita emissions.
Germany / Netherlands	Higher fossil fuel share	**Higher than France / UK (source brief comparison)**	Prosperity alone does not determine outcomes; energy mix matters.

~4 t

France per capita CO_2

~4 t

UK per capita CO_2

The source brief identifies France and the UK as lower per capita than Germany or the Netherlands largely because of energy mix.

Source: Global Carbon Budget 2025; Our World in Data.

KEY TAKEAWAYS

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

01 Global emissions exceed 35 Bt and have not yet peaked.

02 China alone emits over 25% of the global total (~12.5 Bt).

03 Per capita inequalities remain extreme: a 150× gap.

04 US and Europe are declining, but historical cumulative responsibility remains large.

05 Policy and energy mix choices — not just prosperity — determine per capita outcomes.

Negotiation lens: total emissions define near-term mitigation scale; per capita and historical patterns frame equity and responsibility.

Sources: Global Carbon Budget 2025; NASA; NOAA Mauna Loa